



Research article

Health prediction of partially observable failing systems under varying environments

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ABSTRACT

The modern engineering systems often operate under varying environments and only partial information can be observed at discrete monitoring epochs. For such systems, few works have been done for the prognostics of health status using the available environment and monitoring information. Therefore, the aim of this article is to present a new health prediction method for modern engineering systems whose condition is partially observable under varying environments. A dynamic Gamma process is proposed to model the system degradation observations under changing environments. To describe the relation of system actual status to the observed information, a proportional hazard (PH) model integrating internal aging and external observations is presented for modeling the system hazard rate. To realize prediction of residual life of such systems, a matrix operation-based prognostic method is presented to calculate the closed-form solutions of health characteristics for the system. A case study of partially observable failing systems is demonstrated, and comparisons with other recent developed approaches are also given to show the effectiveness of the model.

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1. Introduction

Due to the low fault tolerance required by modern engineering systems such as nuclear power and aerospace industry, it is being more and more important to make accurate reliability estimation for preventing serious impacts of sudden failures on the engineering systems [1,2]. The engineering systems often operate under varying environments for satisfying the diverse operation requirements, which inevitably results in different deterioration behaviors [3]. For such systems, the true level of deterioration is generally unknown and affected by aging, cumulative damage, operating environments, and so on, therefore, effective health prediction of such engineering systems is particularly essential and challenging [4].

Current health prediction methods can be mainly classified as two categories: physics-based approaches and data-driven approaches [5]. The physics-based approaches are often used to capture the mechanics behind failure modes for engineering components. They require complete understanding of the physical theories of deteriorating process of the components. Data-driven

approaches are suitable to be utilized for prognostics of engineering systems whose physical mechanism of failure process is not clear or cannot be accurately described [6,7]. These approaches fit condition monitoring data via artificial intelligence or statistic-based models for predicting future health status and do not rely on the physical principle of the systems [8]. This category of the method shows great potential to exploit the statistical characteristics and is widely used in health assessment for modern engineering systems [9].

In prognostics applications, the condition monitoring data cannot completely identify the actual degradation state of the modern engineering system, which is referred to as the partially observable system [10]. The collected sensor measurements are indirectly and randomly related to the real state of the system health, such as the cutting-force of cutting tools [11], oil wear of transmission units [12], and vibration signal of the pump machinery [13]. For partially observable systems, the data-driven approach typically applies the following statistic-based models for health estimation: stochastic filtering-based model, proportional hazard (PH) model, hidden Markov model (HMM), and hidden semi-Markov model (HSMM). Among them, the PH model has been considered a flexible and powerful tool to integrate multiple degradation features into system failure modeling [14]. PH models use a baseline hazard function to describe the effect of aging and allows the covariate information to represent the accumulation process of partially observable degradation [15]. For instance, Tang et al. [16] focused on the semi-parameter PH

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model and considered the time-independent covariates that have significant impacts on the cable failures. The result shows that Cox's PH model is more adaptive and robust in failure analysis than the Weibull distribution. Hu et al. [17] proposed a Weibull PH model for joint consideration of both degradation and failure data. The closed-form expressions of reliability characteristics are optimized and applied to predictive maintenance of lead–acid batteries. Liu et al. [18] used a PH model to describe the failure rate of a partially observable system that is subject to aging and cumulative damage. The approach was illustrated by a case study of asphalt plug joint in a bridge system. Duan et al. [19] proposed a Gamma covariate PH model for partially observable systems and provided an approximation method to predict health measures of the feed systems, which broadens the application area of PH models.

Even though the PH models have been widely applied in the reliability field, there are some limitations for real applications [20]. For example, the influence of environments was not considered in degradation modeling. In actual operations, the degradation of engineering machines is partially observed and affected by different operational loads and temperatures, which makes reliability prediction quite challenging [21]. To overcome this challenging, some research works considering influence of environments were developed. Peng et al. [22] presented an inverse Gaussian process model for describing the deterioration processes with both constant and time-varying degradation rates. The model was demonstrated via a degradation case of machining tool spindle systems. Shen et al. [23] used a continuous-time Markov process to model the evolution of varying environments. The system degradation is governed by different homogeneous Gamma processes depending on the varying environments. Yang et al. [24] incorporated two random environmental impacts into the competing failures modeling and presented a case study of oil pipeline to illustrate the applicability of their proposed approach. Despite the above-mentioned works have achieved certain levels of success in fully observable systems prognostics, few works consider the influence of varying environments for health prediction of partially observable engineering systems.

PH models are usually employed for describing partial degradation with dynamic factors. Existing PH model-based prognostic works assume that the degradation process is constant within monitoring intervals, and degradation increments are only considered at the monitoring points for reliability estimation. This assumption makes mathematical computation feasible, but is obviously inconsistent with the real deterioration situation, and thus, may decrease the accuracy of health prediction. In the reliability literature, the method developed by Wu et al. [25] provided a recursive procedure for considering non-constant degradation within monitoring intervals. However, only limited degradation states can be addressed with the presented PH model. These limitations hinder the application of the PH model and make the fitting process challenging for prognostic modeling and calculation. Up to now, general approaches for prognostics of partially observable systems under varying environments have not appeared yet.

Motivated by these factors above, in this paper we concentrate on partially observable failing systems and propose a dynamic Gamma process to model the degradation states of the systems under varying environments. The system age, degradation state, and environment profile are jointly considered by a novel PH model with a flexible transition mechanism. A matrix operation-based method is proposed to facilitate the derivation process and provide closed-form expressions of health characteristics including conditional reliability function and mean residual life. To our knowledge, this is a very new work considering partially observable degradation process affected by varying environments. The

purpose of this work is to provide suitable degradation model and health quantity estimates to partially observable systems under operational conditions. The proposed model can be used for a variety of engineering systems such as wind turbines and machine tools under varying environments.

The contributions of this paper are summarized as follows: (1) A dynamic Gamma process is presented for modeling partially observable degradation process under varying environments. (2) A novel PH model with a flexible transition mechanism is developed to describe the relation of observed degradation process to system failure rate. (3) A matrix operation-based prognostic method is presented to calculate health characteristics of partially observable systems. (4) Better prognostic performance is obtained than the models considering constant environment.

The remainder of the paper is organized as follows. In Section 2, we formulate a PH model for partially observable systems under different environment conditions. A dynamic Gamma process is presented for modeling the environment-varying degradation process. In Section 3, we present a matrix operation-based method to realize the estimation of system health quantities. The non-constant transition model under varying environment is considered. In Section 4, an illustrative example is presented to show the effectiveness of the proposed approach, and comparisons are given as well. Finally, conclusions and future works are given in Section 5.

2. Model formulation

Considering an engineering system is under varying operational environments and subject to condition monitoring at discrete time points $(\Delta, 2\Delta, \dots)$, the partial degradation measurements are observed at these regular epochs. The observed degradation process can be affected by the varying environments including the internal loads, ambient temperatures and so on. For example, the degradation rate of wind turbines may change due to varying wind speeds as well as uncertain ambient temperatures. The environmental information (e.g., loads or temperatures) is continuously recorded and can be considered a stochastic process.

For the system operating under different degrees of environment condition, the degradation rate of system is determined by the operational time as well as the impact of varying environments. We thus consider all these processes into degradation modeling. Fig. 1a shows the degradation evolution under constant environment condition, the degradation paths evolve gently, and the degradation rate has subtle changes. Fig. 1b shows the degradation behavior under dynamic environment condition, which is closer to practical cases. The degradation rate changes with operational time and evolves more precipitously due to the varying environment, which indicates the environment state has affected the degradation behavior of the system. Compared to the existing works which focus mainly on the degrading system under constant environment, we intend to investigate and present a general degradation model. In our model, the degradation rate of the system is linked to the usage time t and environment state U_t for $t \geq 0$, which has the following form:

$$\eta(t, U_t) = Z(t) \cdot U_t. \quad (1)$$

where $Z(t)$ is a time-dependent coefficient representing the non-linear behavior of aging effect, and U_t is the environment state at usage time t which is governed by a continuous-time Markov chain with state space $\{1, 2, \dots, D\}$. The certain form of the $Z(t)$ can be higher-order polynomial terms, logarithm function, or power function based on the physical knowledge of degrading systems. The proposed model is suitable for the system that degrades continuously and gradually under varying environments,

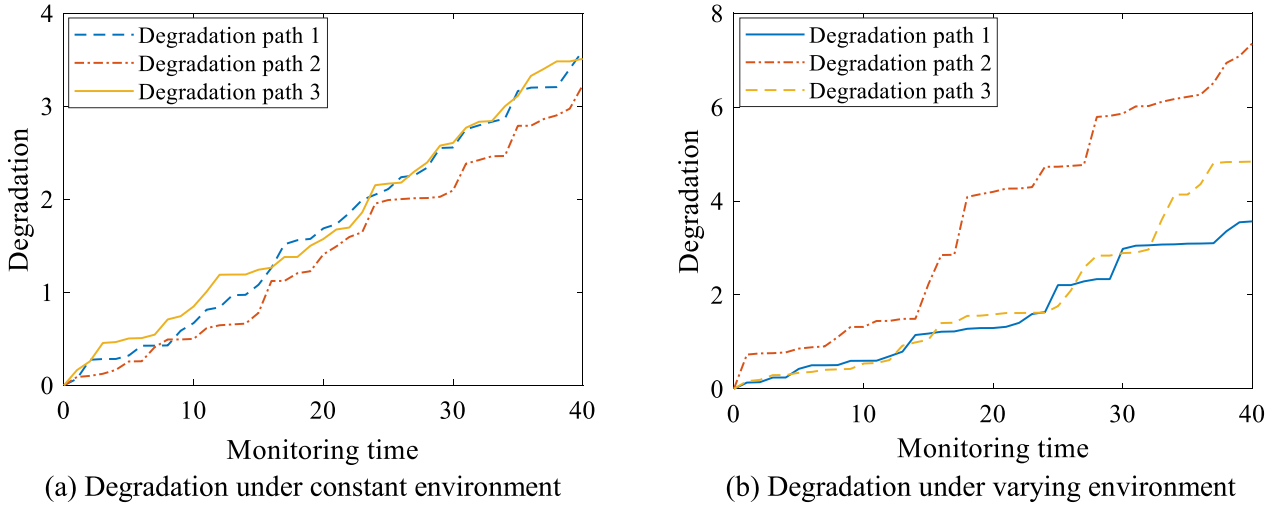


Fig. 1. Illustration of degradation paths.

such as wind turbines. In what follows, we formulate a new advanced degradation process for engineering systems under varying environments.

For most engineering systems, the system degradation can be seen as independent and nonnegative increments for any time interval, such as wear, erosion, and crack growth [26,27]. This characteristic is particularly suitable for Gamma process whose degradation for any time interval can be considered as increments of small wear or shock incurred. We assume that the system degradation process under the varying environment $\{X(t, U_t), t \geq 0\}$ starts from a known initial state X_0 and follows a dynamic Gamma process. Hence, the probability density function of the degradation process $X(t, U_t)$ has the following form:

$$\begin{aligned} f_{X(t, U_t)}(x) &= Ga(x|\eta(t, U_t), v) \\ &= \frac{v^{\eta(t, U_t)}}{\Gamma(\eta(t, U_t))} x^{\eta(t, U_t)-1} \exp(-vx). \end{aligned} \quad (2)$$

where $\eta(t, U_t)$ is the shape parameter affected by the environment-varying degradation. v is scale parameter. $\Gamma(\eta(t, U_t)) = \int_0^\infty u^{\eta(t, U_t)-1} \exp(-u) du$ is the gamma function. Note that the shape parameter $\eta(t, U_t)$ depends on both usage time and environment state, which signifies that the aging and environment condition are incorporated into the degradation paths.

Due to the infinite divisibility property of Gamma process, the degradation increments in any small-time intervals follow Gamma distributions. We define a small value δ as the resolution of the monitoring interval Δ . For any small δ , the developed degradation model has following characteristics:

- For any time points $0 < \delta < 2\delta < \dots < \omega\delta < \infty$, the degradation increments $X(\delta, U_\delta) - X_0$, $X(2\delta, U_{2\delta}) - X(\delta, U_\delta)$, $\dots$, $X(\omega\delta, U_{\omega\delta}) - X(\omega\delta - \delta, U_{\omega\delta - \delta})$ are mutually independent.
- For any intervals $[\omega\delta - \delta, \omega\delta]$, the degradation increments $X(\omega\delta, U_{\omega\delta}) - X(\omega\delta - \delta, U_{\omega\delta - \delta})$ are distributed as $Ga(\eta(\omega\delta, U_{\omega\delta}) - \eta(\omega\delta - \delta, U_{\omega\delta - \delta}), v)$.

The observed degradation is partially related to the system health status and will lead to random failure eventually. To incorporate the observed degradation signal into the random failure model, we present a new PH model to characterize the failure rate of the system as follows:

$$h(t, X(t, U_t)) = h_0(t) \cdot \psi(X(t, U_t); \beta). \quad (3)$$

where $h_0(t)$ is the baseline hazard rate function at time t . $\psi(\cdot)$ is a covariate which represents the effect of degradation process affected by usage and environment state (i.e., loads, temperatures

and so on). β is the regression parameter depending only on the values of the degradation $X(t, U_t)$. For system operating at time $\omega\delta$, the failure rate can be expressed as

$$h(\omega\delta, X(\omega\delta, U_{\omega\delta})) = h_0(\omega\delta) \cdot \psi(X(\omega\delta, U_{\omega\delta}); \beta). \quad (4)$$

Based on the formulated model, in this work we aim to derive the health characteristics, including system conditional reliability, mean residual lifetime, and residual life distribution. However, for such a degrading system, the health characteristics estimation is quite challenging since the degradation path $X(\omega\delta, U_{\omega\delta})$ is an environment-varying process. Also, the degradation information is only known at monitoring points, which make computation extremely complicated. To overcome this challenge, an approximation technique is proposed for health characteristics prognostics in the next section.

3. Health characteristics prognostics

To estimate the health characteristics of the engineering systems, the non-monitored degradation and the environment condition should be considered. Therefore, we first calculate the degradation transition probability for every single δ . For time point $\delta, 2\delta, \dots, N\delta$ ($N\delta = \bar{T}$ is predetermined life limit), the conditional reliability that system will survive before time T given the current observation at environment state $U_{\omega\delta}$ and usage time $\omega\delta$ can be expressed as

$$\begin{aligned} R(T|X(\omega\delta, U_{\omega\delta}), \omega\delta, U_{\omega\delta}) &= \Pr(\xi > T | \xi > \omega\delta, X(\omega\delta, U_{\omega\delta}), \omega\delta > T) \\ &= \exp\left(-\int_{\omega\delta}^T h(t, X(t, U_t)) dt\right). \end{aligned} \quad (5)$$

where ξ is the system failure time, and $X(\omega\delta, U_{\omega\delta})$ is the degradation level at approximation time $\omega\delta$ for $0 \leq \omega \leq N$. However, due to the non-stationary characteristic and integrated varying environments in Eq. (5), it is quite tricky or even impossible to derive the explicit formula of health characteristics.

To address this difficulty, an effective approximation method based on the discrete-state process inspired by Brook and Evans [28] is proposed. We discretize the range of system degradation process $[0, z]$ into M states with equal interval d , and the state space of degradation process is defined as $\Omega = \{0, 1, \dots, M\}$, where state 0 denotes the initial new state, and M denotes the state above the degradation level z . The degradation level can transit from state i to any higher state j or stay in the same

state for a very short time interval, $i, j \in \Omega$. The evolution of environment state is driven by a continuous-time Markov chain with transition rate λ_{kl} and transition probability P_{kl} for $k, l \in \{1, 2, \dots, D\}$. To jointly record the degradation level, usage time, and varying environment, we define the overall system state as $(i, \omega\delta, k)$ given degradation level $X(\omega\delta, k) = i$ at usage time $\omega\delta$ and environment state k . The one-step transition probability of system state within an approximation interval δ can be expressed as

$$\begin{aligned} & \Phi_{ij,kl}(\omega\delta) \\ &= \Pr(X(\omega\delta + \delta, l) = j, \xi > \omega\delta + \delta, U_{\omega\delta+\delta} = l | X(\omega\delta, k) \\ &= i, \xi > \omega\delta, U_{\omega\delta} = k) \\ &= \Pr(X(\omega\delta + \delta, l) = j, U_{\omega\delta+\delta} = l | X(\omega\delta, k) \\ &= i, \xi > \omega\delta + \delta, U_{\omega\delta} = k) \\ &\cdot \Pr(\xi > \omega\delta + \delta | X(\omega\delta, k) = i, \xi > \omega\delta, U_{\omega\delta} = k). \end{aligned} \quad (6)$$

The second term in (6) represents the system conditional reliability at state $(i, \omega\delta, k)$. The first term in (6) represents the probability that the system has not failed, and the degradation transits to degradation state j and environment state l after a time interval δ . For a small interval $T = \delta$, the system conditional reliability in (5) can be further computed as

$$\begin{aligned} & R(\omega\delta + \delta | X(\omega\delta, U_{\omega\delta}), \omega\delta, k) \\ &= \sum_{j=0}^M \left(\begin{aligned} & \Pr(X(\omega\delta + \delta, l) = j, U_{\omega\delta+\delta} = l | X(\omega\delta, k) \\ &= i, \xi > \omega\delta + \delta, U_{\omega\delta} = k) \\ & \cdot \Pr(\xi > \omega\delta + \delta | X(\omega\delta, k) = i, \xi > \omega\delta, U_{\omega\delta} = k) \end{aligned} \right) \quad (7) \\ &= \sum_{l=1}^D \sum_{j=0}^M \Phi_{ij,kl}(\omega\delta). \end{aligned}$$

To compute the conditional reliability in (7), it is required to derive the first term in (6) for each possible degradation state under an environment state for a small interval δ . Each transition occurring in a small interval should be considered. Therefore, for system operating with a small interval δ , there are four scenarios of one-step transition probabilities for the first term in (6).

Suppose that the possible transition occurs at time $u + \omega\delta$, $u + \omega\delta \in (\omega\delta, \omega\delta + \delta)$, the first term in (6) are derived as follows.

• When $i = j$, $i, j \in \Omega$ and $k = l$, $k, l \in \{1, 2, \dots, D\}$, the probability term can be derived as:

$$\begin{aligned} & \Pr(X(\omega\delta + \delta, l) = j, U_{\omega\delta+\delta} = l | X(\omega\delta, k) = i, \xi > \omega\delta + \delta, U_{\omega\delta} = k) \\ &= \int_{\delta}^{+\infty} \int_0^{\frac{\delta}{2}} Ga(\Delta\eta(\omega\delta, k), v) dx \cdot \lambda_k \cdot \exp(-\lambda_k u) du. \end{aligned} \quad (8)$$

where λ_k is rate parameter of the sojourn time for environment state k , and $\Delta\eta(\omega\delta, k) = \eta(\omega\delta + \delta, k) - \eta(\omega\delta, k)$.

• When $i < j$, $i, j \in \Omega$ and $k = l$, $k, l \in \{1, 2, \dots, D\}$, the probability term can be derived as:

$$\begin{aligned} & \Pr(X(\omega\delta + \delta, l) = j, U_{\omega\delta+\delta} = l | X(\omega\delta, k) = i, \xi > \omega\delta + \delta, U_{\omega\delta} = k) \\ &= \int_{\delta}^{+\infty} \int_{(j-i)\delta-\frac{\delta}{2}}^{(j-i)\delta+\frac{\delta}{2}} Ga(\Delta\eta(\omega\delta, k), v) dx \cdot \lambda_k \cdot \exp(-\lambda_k u) du. \end{aligned} \quad (9)$$

• When $i = j$, $i, j \in \Omega$ and $k \neq l$, $k, l \in \{1, 2, \dots, D\}$, the probability term can be derived as:

$$\begin{aligned} & \Pr(X(\omega\delta + \delta, l) = j, U_{\omega\delta+\delta} = l | X(\omega\delta, k) = i, \xi > \omega\delta + \delta, U_{\omega\delta} = k) \\ &= P_{kl} \int_0^{\delta} \int_0^{\frac{\delta}{2}} Ga(\Delta\eta(\omega\delta, (k+l)/2), v) dx \\ &\cdot \lambda_k \cdot \exp(-\lambda_k u) \cdot \exp(-\lambda_l(\delta - u)) du. \end{aligned} \quad (10)$$

where P_{kl} is the transition probability from environment state k to environment state l .

• When $i < j$, $i, j \in \Omega$ and $k \neq l$, $k, l \in \{1, 2, \dots, D\}$, the probability term can be derived as:

$$\begin{aligned} & \Pr(X(\omega\delta + \delta, l) = j, U_{\omega\delta+\delta} = l | X(\omega\delta, k) = i, \xi > \omega\delta + \delta, U_{\omega\delta} = k) \\ &= P_{kl} \cdot \int_0^{\delta} [(G_1 + G_2) \cdot \lambda_k \cdot \exp(-\lambda_k u - \lambda_l(\delta - u))] du. \end{aligned} \quad (11)$$

where the notations in (11) are $G_1 = \int_{(j-i)\delta-\frac{\delta}{2}}^{(j-i)\delta+\frac{\delta}{2}} Ga(\eta(\omega\delta + u, k) - \eta(\omega\delta, k), v) dx$ and $G_2 = \int_{(j-i)\delta}^{(j-i)\delta+\frac{\delta}{2}} Ga(\eta(\omega\delta + \delta, l) - \eta(\omega\delta + u, l), v) dx$.

Finally, the one-step transition probability matrix for interval δ can be obtained as $\Phi(\omega\delta) = (\Phi_{ij,kl}(\omega\delta))$ for $i, j \in \Omega$, $k, l \in \{1, 2, \dots, D\}$, $0 \leq \omega \leq N$. If the system failure state is denoted by F state, then the probability that system transits from state $(i, \omega\delta, k)$ to state F during interval δ can be expressed as

$$P_{ik,F}(\omega\delta) = 1 - \sum_{l=1}^D \sum_{j=0}^M \Phi_{ij,kl}(\omega\delta). \quad (12)$$

From (8)–(12), we can calculate the system reliability characteristic for a small interval δ . However, for a longer interval, we consider the time duration is composed of multiple δ . To jointly consider the multiple successive δ , we present a computation process in matrix form, which takes advantage of property of matrix operation. Hence, we formulate a matrix containing all transition probabilities occurring in each time interval δ of the entire life of systems, which is given by

$$\mathbf{P} = \begin{bmatrix} 0 & \Phi(0) & 0 & \dots & 0 & \dots & 0 & 1 - \Phi(0)e_{DM} \\ 0 & 0 & \Phi(\delta) & \dots & 0 & \dots & 0 & 1 - \Phi(\delta)e_{DM} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \vdots \\ 0 & 0 & 0 & \dots & \Phi(\omega\delta) & \dots & 0 & 1 - \Phi(\omega\delta)e_{DM} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \vdots \\ 0 & 0 & 0 & \dots & 0 & \dots & \Phi((N-2)\delta) & 1 - \Phi((N-2)\delta)e_{DM} \\ 0 & 0 & 0 & \dots & 0 & \dots & \Phi((N-1)\delta) & 1 - \Phi((N-1)\delta)e_{DM} \\ 0 & 0 & 0 & \dots & 0 & \dots & 0 & 1 \end{bmatrix} \quad (13)$$

In the matrix $\mathbf{P}$, e_{DM} is an identity vector of $D \times M$ units, 0 is $DM \times DM$ zero matrix, 1 is a unit matrix, and $1 - \Phi(\omega\delta)e_{DM}$ is the probability that system transitions to the failure state during the next interval δ .

To simplify the matrix $\mathbf{P}$ and make the calculation feasible, a matrix operation-based method is presented to partition the matrix $\mathbf{P}$ into the following form:

$$\mathbf{P} = \begin{bmatrix} \mathbf{B} & \mathbf{P}_F \\ 0^T & 1 \end{bmatrix} \quad (14)$$

where $\mathbf{B}$ is a matrix of $DMN \times DMN$ dimensions, and $\mathbf{P}_F$ is a vector of DMN elements.

Instead of calculating the health quantities for a time interval with one possible transition, we have to compute the reliability characteristic for a longer interval with multiple transitions. Through r step of multiplication of matrix $\mathbf{P}$, all the possible transitions within $r\delta$ are considered. The obtained matrix is given

by

$$\mathbf{P}^r = \begin{bmatrix} \mathbf{B}^r & \mathbf{1} - \mathbf{B}^r \\ \mathbf{0} & 1 \end{bmatrix}, \quad (r = 1, 2, \dots) \quad (15)$$

where $\mathbf{1}$ is a unit matrix of $DMN \times DMN$ dimensions. We notice that the r th position from the right of the main block diagonal of $\mathbf{B}^r$ has the following form: $\Phi(\omega\delta) \cdot \Phi((\omega+1)\delta) \cdots \Phi((\omega+r-1)\delta)$, which indicates all transition probabilities of the system state during time interval $r\delta$. Based on the block matrix $\mathbf{B}^r$, various health characteristics can be obtained easily. Referring to Eqs. (5) and (7), for a long survival time $r\delta$, the conditional reliability for the system state $(i, \omega\delta, k)$ is given by

$$\begin{aligned} R((\omega+r)\delta|i, \omega\delta, k) \\ = 1 - \sum_{t=1}^r \Pr(RL = t\delta | X(\omega\delta, k) = i, \xi > \omega\delta, U_{\omega\delta} = k) \\ = 1 - \pi_{i,k}(\omega\delta)(\mathbf{1} - \mathbf{B}^r)\mathbf{e}_{DMN}. \end{aligned} \quad (16)$$

where $r = 1, \dots, N - \omega$. The row probability vector $\pi_{i,k}(\omega\delta) = (0, \dots, 0, 1, 0, \dots, 0)$ is $1 \times DMN$ with 1 for $(\omega DM + iD + k)$ th element, and zeros elsewhere. $\mathbf{e}_{DMN}$ is an identity vector with DMN units.

Besides conditional reliability function, the residual lifetime (RL) is also an important characteristic which is widely considered in health prediction. Define $RL_{i,\omega,k}$ as the residual lifetime of the system at state $(i, \omega\delta, k)$. The cumulative distribution function

of the failure occurrence can be obtained as:

$$\begin{aligned} \Pr(\xi \leq (\omega+r)\delta | X(\omega\delta, k) = i, \xi > \omega\delta, k) \\ = \pi_{i,k}(\omega\delta)(\mathbf{1} - \mathbf{B}^r)\mathbf{e}_{DMN}. \end{aligned} \quad (17)$$

From the partitioned matrix, the probability mass function of $RL_{i,\omega,k}$ for the system state $(i, \omega\delta, k)$ is given by

$$\begin{aligned} \Pr(RL_{i,\omega,k} = r\delta | X(\omega\delta, k) = i, \xi > \omega\delta, k) \\ = \pi_{i,k}(\omega\delta)\mathbf{B}^{r-1}(\mathbf{1} - \mathbf{B})\mathbf{e}_{DMN}. \end{aligned} \quad (18)$$

Based on the probability mass function, we calculate the mean residual life for system state $(i, \omega\delta, k)$ as

$$\begin{aligned} \text{MRL}_{i,\omega,k} = \sum_{r=1}^{N-\omega} \Pr(RL_{i,\omega,k} = r\delta | X(\omega\delta, k) = i, \xi > \omega\delta, k) \cdot r\delta \\ = \delta \pi_{i,k}(\omega\delta)(\mathbf{1} - \mathbf{B})^{-1}\mathbf{e}_{DMN}. \end{aligned} \quad (19)$$

From Eqs. (17)–(19), we can see that the system health characteristics are explicitly expressed by formulas of matrix $\mathbf{B}$, which can be obtained via several steps of matrix operations with PH model parameter estimates. In the next section, we will illustrate the application process of the proposed prediction model and compare the proposed approach with the model that does not take the varying environments into account.

4. Application illustration

4.1. Case study

The purpose of this section is to show how the proposed approach can be applied to an engineering system through an

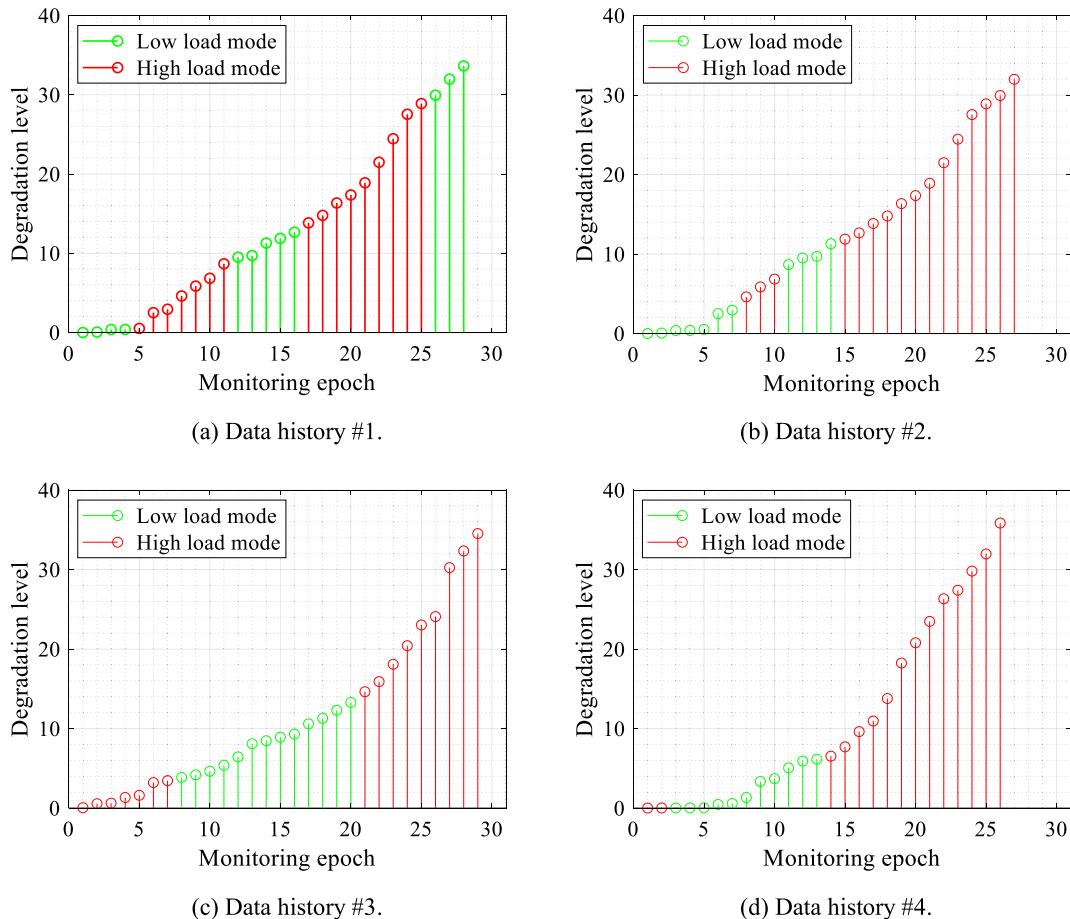
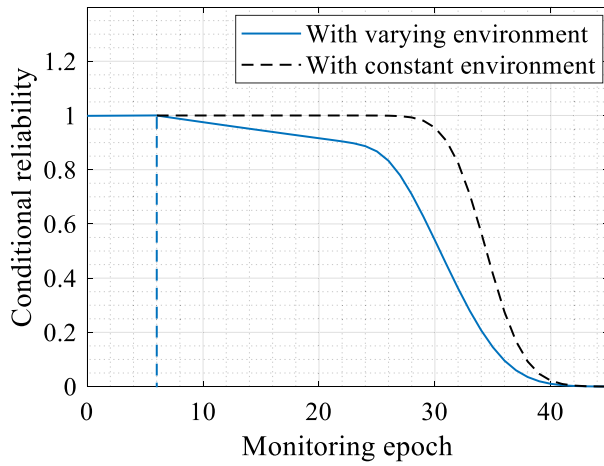
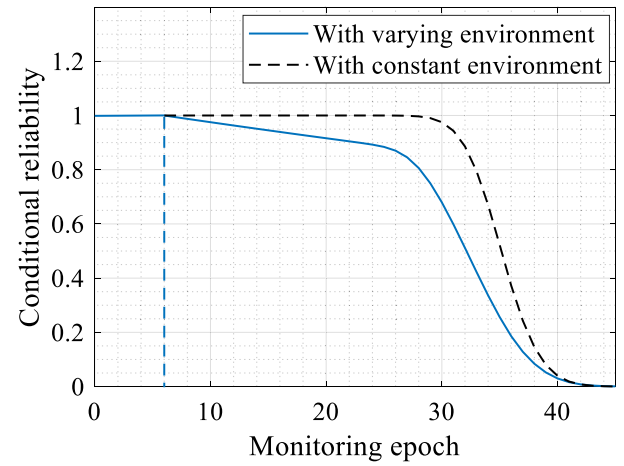


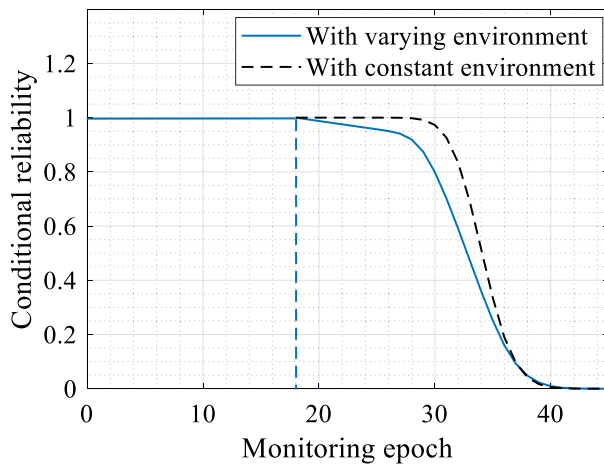
Fig. 2. Illustration of selected failure histories for wind turbine gearboxes.



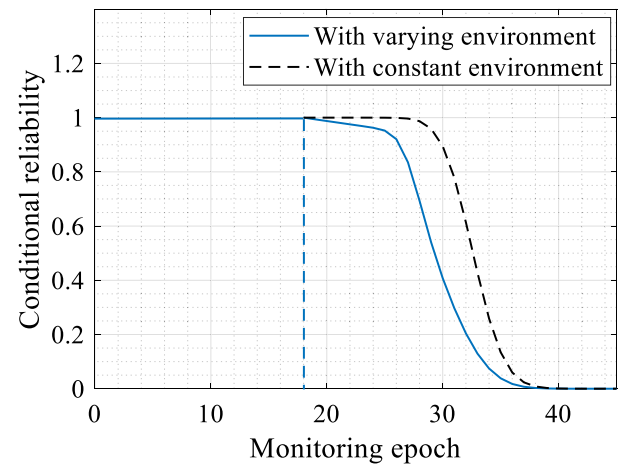
(a) Data history #1.



(b) Data history #2.



(c) Data history #3.



(d) Data history #4.

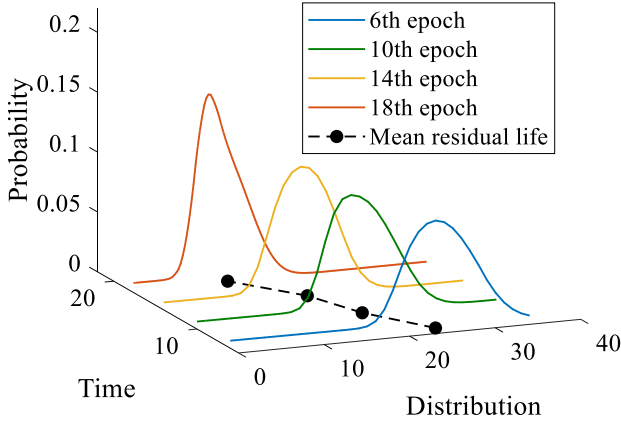
Fig. 3. Predicted conditional reliability of gearbox systems.

example study of a wind turbine under varying environments. Wind turbines are widely used in energy field and suffer different wind speeds, which lead to different operational conditions of the gearbox system. The wind speeds are classified as high if the speed is equal or above 10 m/s and as low if the speed is below 10 m/s. The two environment states correspond to the high load mode and low load mode of the wind turbine gearbox system. The wind turbine may fail from any degradation state at any operational time [29–31]. During the condition monitoring of the gearbox system, the wear particles of lubricating oil are collected at each $\Delta = 5$ h and measured by ferrographic analysis. Two types of data histories are obtained, including 20 data histories, which 6 of them are failure histories and 14 of them are censored histories. The failure history is the observation that ends with a failure, while the censored history is the observation that ends with suspension of maintenance. Through the data examination, the wear accumulation itself does not directly result in system failure but increases the risk of failure. The usage time also leads to the increase of failure probability of the wind turbines. Therefore, we describe the failure rate using a PH model to consider aging process by baseline hazard function and environment-varying degradation process by covariates.

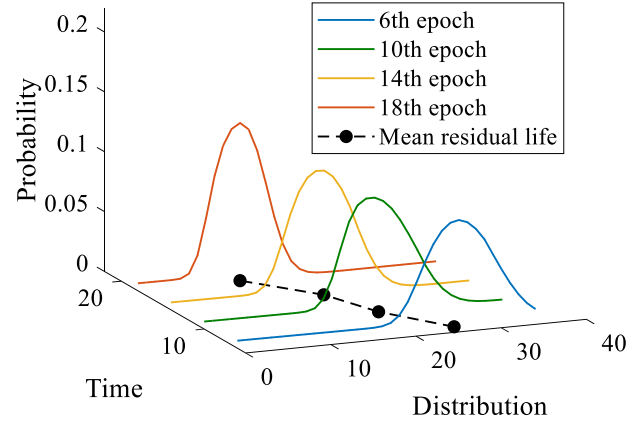
The predetermined life limit of the gearbox system is $\bar{T} = 175$ h. Fig. 2 shows four selected failure histories #1, #2, #3, and #4 with corresponding failure times 140 h, 135 h, 145 h, and 130 h, respectively. In Fig. 2a, 2b, 2c, and 2d, the green color denotes that the wind turbine operates under low load environment mode while the red one represents the high load environment mode. We denote them as state 1 and state 2 correspondingly for the following investigations.

To fit the proposed PH model, we conduct the parameter estimation procedure with two steps. First, we estimate the parameters of the PH model using censored histories and failure histories. Then, we estimate parameters of dynamic Gamma process using observation sequences. The procedure is implemented as follows. For L independently observed data histories, let $(T_p, C_p, X(t, U_t); t \leq T_p)$ denote the p th, $p \in \{1, 2, \dots, L\}$ observed data history, which may be ended either by failure occurrences or by censoring. The censoring indicator C_p is defined as

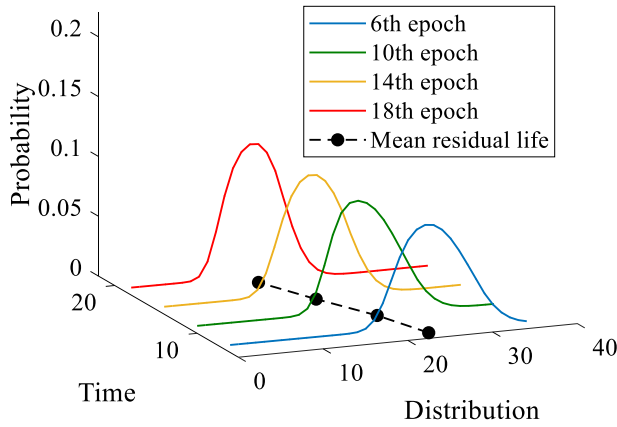
$$C_p = \begin{cases} 1 & \text{if the failure occurs,} \\ 0 & \text{if the system is censored.} \end{cases} \quad (20)$$



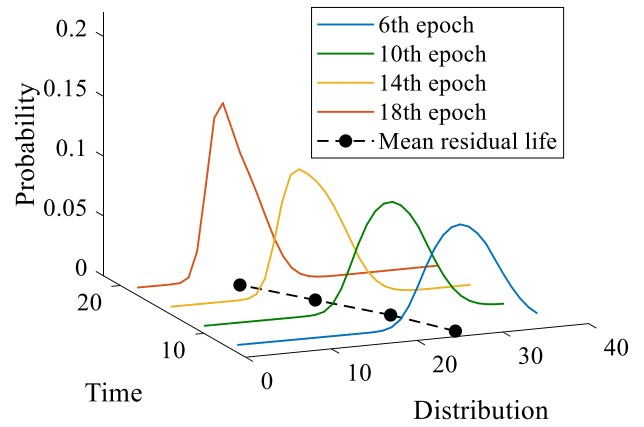
(a) Data history #1.



(b) Data history #2.



(c) Data history #3.



(d) Data history #4.

Fig. 4. Predicted residual life distribution of gearbox systems.

where T_p denotes the lifetime of the p th history. Let the p th history be monitored at $t_p = (t_{p,0} < t_{p,1} < \dots < t_{p,k_p})$ and corresponding degradation observations be $X_p = (x_{p,0}, x_{p,1}, \dots, x_{p,k_p})$.

The baseline distribution of PH model is assumed to be Weibull with the hazard rate function $h_0(t) = \gamma t^{\gamma-1}/\alpha^\gamma$, where α is the scale parameter and γ is the shape parameter. The covariate of PH model is chosen to be exponential function $\psi(X(t); \beta) = \exp(\beta X(t))$. The exponential function is more appropriate to quantify the deterioration effect over the linear function since the growing degradation has an exponential impact on the failure process. The parameters for failure rate function are defined as $\mu = [\alpha, \beta, \gamma]$. The likelihood function can be written as

$$L(\mu) \propto \prod_{p=1}^L f(T_p, X_p; \mu)^{C_p} R(T_p, X_p; \mu)^{1-C_p}. \quad (21)$$

where $f(T_p, X_p; \mu)$ is the failure function, and $R(T_p, X_p; \mu)$ is the reliability function. The log-likelihood function can be written as the following form:

$$\begin{aligned} \ln(L(\mu)) &\propto \sum_{p=1, C_p=1}^L \ln\left(\frac{\gamma t_{p,k_p}^{\gamma-1}}{\alpha^\gamma} \cdot \exp(\beta \cdot x_{p,t_{p,k_p}})\right) \\ &\quad - \sum_{p=1}^L \int_0^{t_{p,k_p}} \exp(\beta \cdot x_s) d(s/\alpha)^\gamma \\ &\approx \sum_{p=1, C_p=1}^L \ln\left(\frac{\gamma t_{p,k_p}^{\gamma-1}}{\alpha^\gamma} \cdot \exp(\beta \cdot x_{p,t_{p,k_p}})\right) \\ &\quad - \sum_{p=1}^L \sum_{n=0}^{k_p-1} \exp(\beta \cdot x_{t_{p,n}}) [(t_{p,n+1}/\alpha)^\gamma - (t_{p,n}/\alpha)^\gamma]. \end{aligned} \quad (22)$$

The parameters $\mu = [\alpha, \gamma, \beta]$ of the PH model are estimated by maximizing the log-likelihood function in Eq. (22). Next, we proceed to estimate the parameters of dynamic Gamma process. The $Z(t) = at^b$ and environment state space is $\{1, 2\}$. The function form is chosen for quantifying the deterioration rate over the linear and logarithm functions since the deterioration slope presents power growth trend over usage time. The environment transition rates λ_{12} and λ_{21} can be estimated by examining the environment records. The unknown parameters of the dynamic

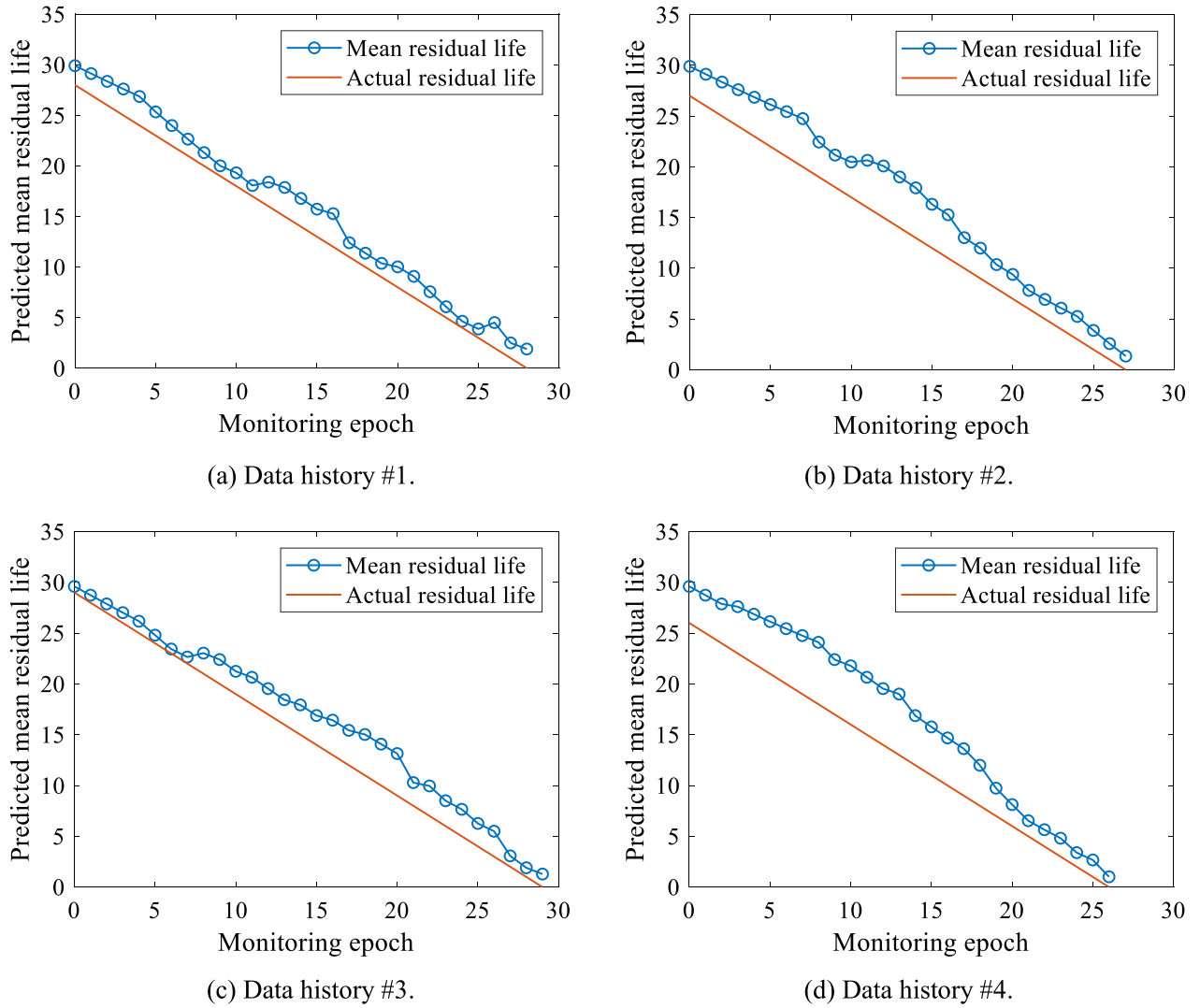


Fig. 5. Predicted mean residual life using the proposed approach.

Gamma process are α, b, v . The likelihood function of the observed degradation increments $\Delta x_{p,i} = x_{p,i} - x_{p,i-1}$, $i = 1, \dots, k_p$, $p = 1, \dots, L$, is a sequence of independent Gamma densities.

$$\begin{aligned}
 & l(\Delta x_{1,1}, \dots, \Delta x_{1,k_1}, \dots, \Delta x_{L,1}, \dots, \Delta x_{L,k_p} | a, b, v) \\
 &= \prod_{p=1}^L \prod_{i=1}^{k_p} \frac{v^{a \cdot U_{p,i}} [t_{p,i}^b - t_{p,i-1}^b]}{\Gamma(a \cdot U_{p,i} [t_{p,i}^b - t_{p,i-1}^b])} \delta_{p,i}^{a \cdot U_{p,i} [t_{p,i}^b - t_{p,i-1}^b] - 1} \exp\{-v \delta_{p,i}\}.
 \end{aligned} \quad (23)$$

The parameter α, b, v can be estimated by maximizing Eq. (23). Following the above procedures, we code (20)–(23) into Matlab software. The estimates of α, γ, β are obtained as 5034.34, 4.35, 0.64, respectively. The transition rates of environment are $\lambda_{12} = 0.1007$ and $\lambda_{21} = 0.1341$. The parameters of degradation process are estimated as $a = 0.055$ and $b = 2$, and the scale parameter is $v = 2.17$.

Using the estimated parameters of the model, the matrix $\mathbf{B}$ is obtained by calculating one-step transition probability elements in (8)–(11). We then calculate the conditional reliability that the system is operational until the next monitoring epoch. The conditional reliability in (7) is estimated with matrix $\mathbf{B}$ and state observations. Using (16), we input the state observations, the

conditional reliability of each state of the investigated histories #1, #2, #3, and #4 is predicted and shown in Fig. 3. The constant environment model, which considers a homogeneous Gamma degradation process, is also presented. In Figs. 3a and 3b, the conditional reliability is predicted at early stage of the degradation (i.e., the 6th monitoring epoch) for both failure histories (#1 and #2). While in Figs. 3c and 3d, conditional reliability is illustrated for the late stage of system deterioration (i.e., 18th monitoring epoch). For both deterioration stages, the conditional reliability considering varying environment is lower than the results under constant environment, which signifies that the consideration of environments leads to less delay of the reliability prognostics. Also, the decreasing in reliability of the system is varying with degradation and environment states. We can notice that when wind turbine gearbox system operates with higher degradation and environment state at the same monitoring points, the corresponding conditional reliability levels decreases much more rapidly (see Figs. 3c and 3d).

Using (18)–(19) in Section 3, the mean residual life of the gearbox system and its residual life distribution at different monitoring epochs are obtained. In Fig. 4, four data histories shows that the residual life is decreasing, and its distribution interval is becoming shorter over system degradation. This signifies that

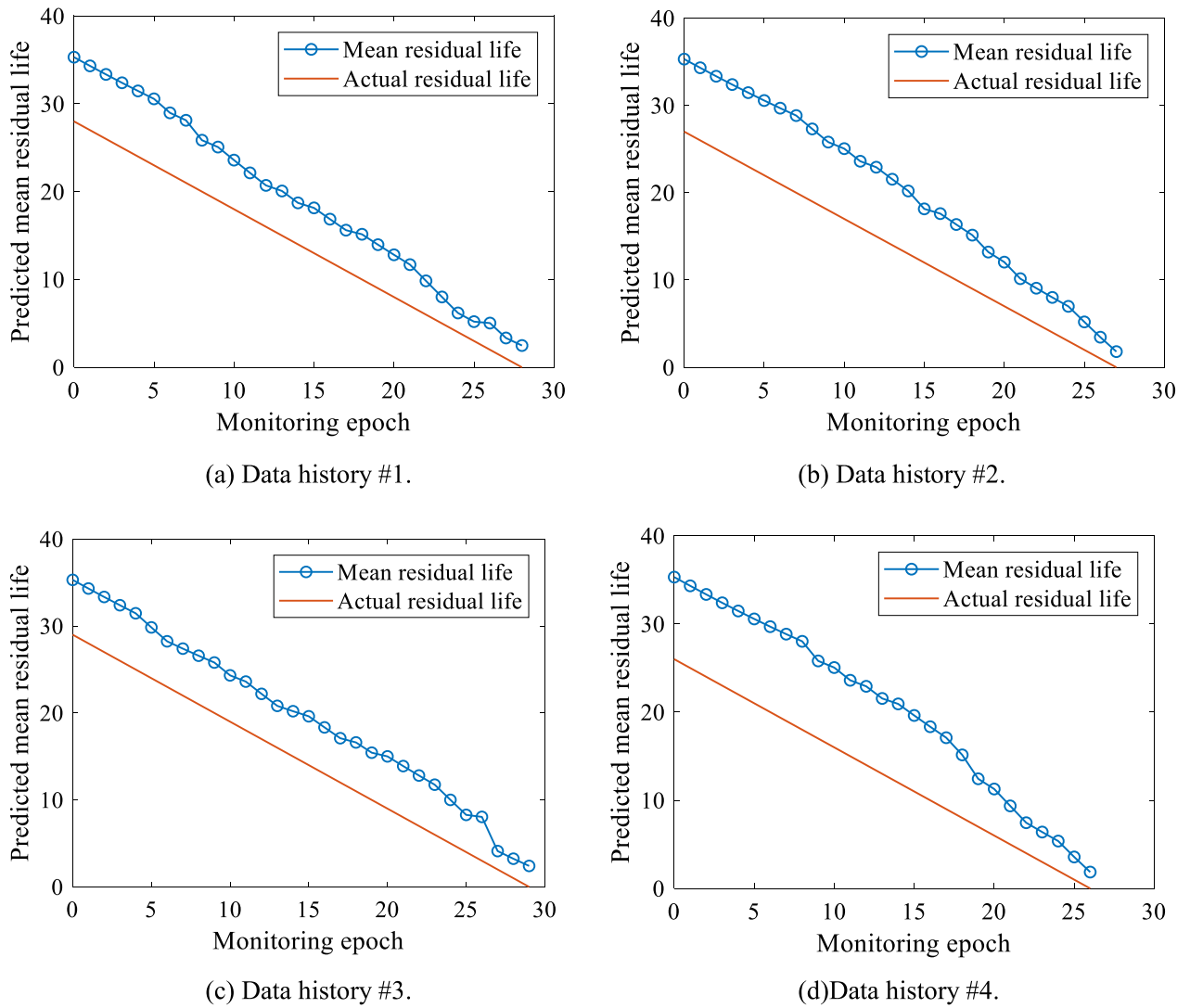


Fig. 6. Predicted mean residual life using the fixed model.

a higher probability of the failure occurrence falls into a smaller range of time over gearbox aging.

Also, the predicted mean residual life considering varying environments for illustrated failure histories #1, #2, #3, and #4 are presented in Fig. 5, which shows the predicted mean residual life for the proposed models as well as the actual residual life. From Figs. 5a–5d, the predicted mean residual lives for four failure histories are very close to the actual residual life, which shows a desirable performance of failure predictions.

For investigating the computational performance, it should be noticed that the matrix $\mathbf{B}$ in (17)–(19) is associated with the parameter estimates of PH model and dynamic Gamma process. This fact signifies that matrix $\mathbf{B}$ can be pre-determined by analyzing data histories, and new prognostics can be obtained by several computation steps through (17)–(19) in real applications. This operation will cost only few seconds by using Matlab software on an Intel Core i5-7300HQ 2.50 GHz, which is very fast and attractive for real-time applications.

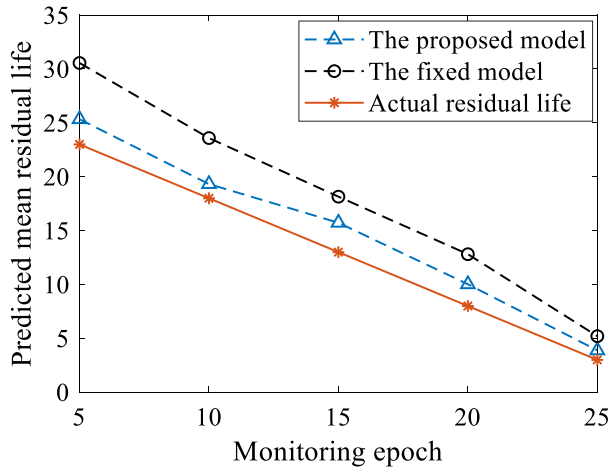
We further study the influence of discretization level on the prognosis result of the proposed model. The predicted mean residual lives under different discretization levels are illustrated in Table 1. When the number of discretization levels exceeds 25, the predicted mean residual life has not changed much, which

means the prediction has a sufficient precision. Therefore, discretization level does not need to be chosen very large.

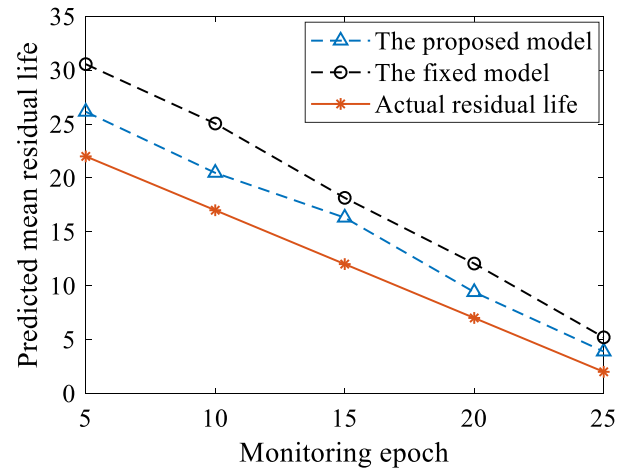
4.2. Comparison to recent PH models

To verify the performance of the proposed approach under varying operating environment, we fit the data histories using PH model considering constant operating condition setting. The contrast PH model is presented by Duan et al. [19] (refer to as the fixed model), in which a homogeneous Gamma process is utilized to describe the degradation process. Fig. 6 shows the predicted mean residual life under constant environment using the fixed model. From Fig. 6, the fixed model using constant environment gives acceptable prediction of mean residual life for failure histories. However, for all the investigated failure histories, Fig. 6 shows the larger prediction delays for the model without considering operational environments.

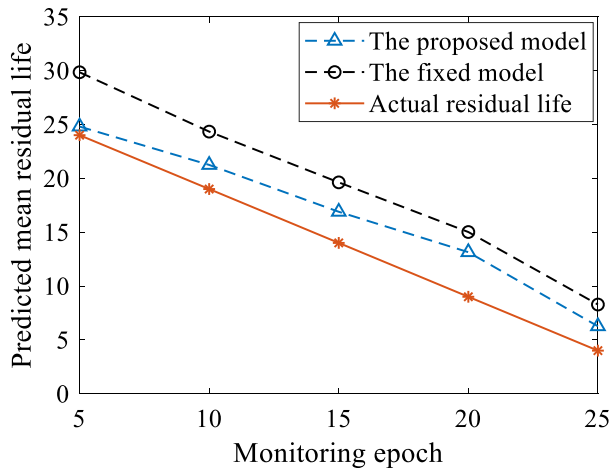
We have compared our proposed approach with the PH model considering constant environment condition setting. Fig. 7 shows the comparison results of mean residual life for both approaches. It is clear to see that both approaches can provide effective mean residual life prediction, but the mean residual life obtained by our model are much closer to the actual residual life than the results



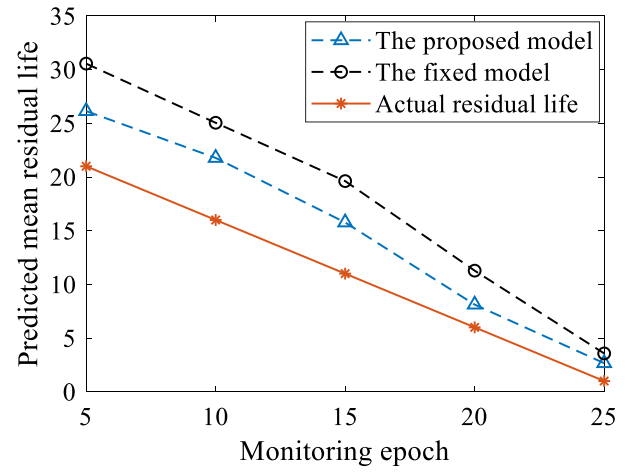
(a) Comparison for data history #1.



(b) Comparison for data history #2.

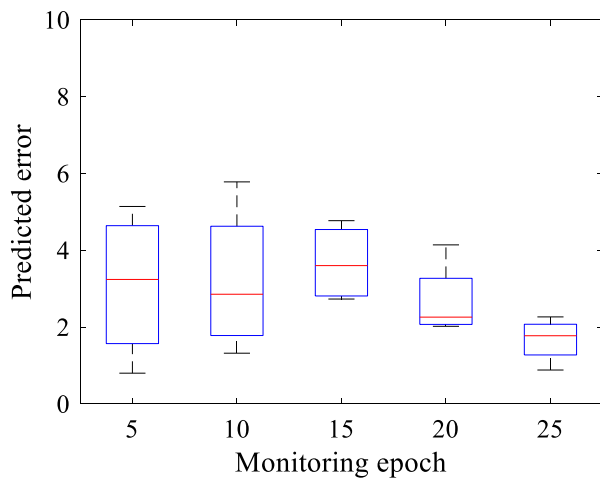


(c) Comparison for data history #3.

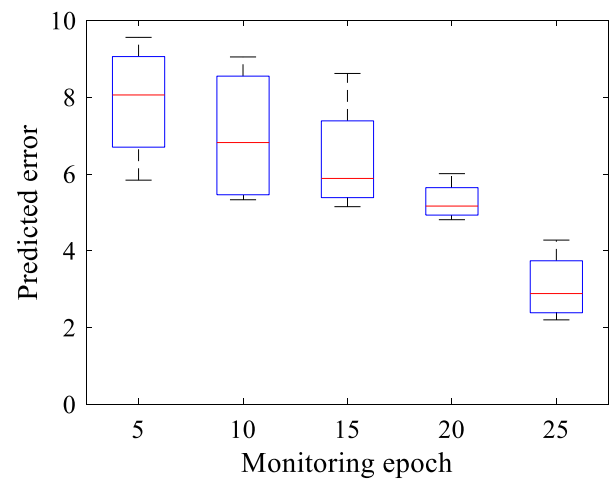


(d) Comparison for data history #4.

Fig. 7. Prediction comparison using two PH approaches.



(a) The approached model.



(b) The fixed model.

Fig. 8. Prediction error using two PH approaches.

Table 1

The effect of discretization level on residual life prediction result for the wind turbine gearbox system.

Discretization level (M)	15	25	35	45
MRL at the 16th epoch	15.3986	14.4758	14.4677	14.4677
MRL at the 26th epoch	1.0843	1.0115	1.0028	1.0028

Table 2

Performance evaluation of three investigated models.

#	Proposed model			Duan's model			Tang's model		
	ME	MSE	MRE (%)	ME	MSE	MRE (%)	ME	MSE	MRE (%)
1	1.90	4.06	14.13	5.14	29.27	22.32	18.89	408.33	32.26
2	3.29	12.09	14.46	6.48	46.39	27.23	19.89	449.62	47.23
3	2.13	5.74	9.19	5.25	28.81	26.98	18.43	401.86	29.36
4	4.01	18.75	11.65	7.44	61.81	36.82	19.82	445.84	34.84

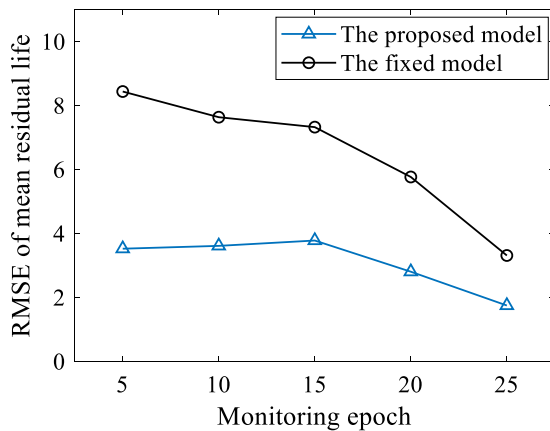


Fig. 9. RMSE of predicted mean residual life for two PH approaches.

obtained by the model without considering operating environment, particularly for failure histories #1 and #4 in Figs. 7a and 7d, respectively.

Moreover, we compare the prediction errors between the mean residual life estimates and actual residual lives for four illustrated failure histories using the two approaches. The prediction errors are the absolute differences from the estimated values to the actual levels of residual life. Fig. 8 provides the prediction errors at different monitoring epochs. The prediction errors are decreased with monitoring time since more monitoring information is available than can enhance the prediction performance. It is also found that model considering dynamic operational conditions provides the lower prediction errors over monitoring time than the constant environment model for investigated data histories.

For further evaluation of the proposed model, we have computed the root mean square error (RMSE) of predicted mean residual life for testing the performance of the proposed model and presented results in Fig. 9. It can be signified that the proposed model gives better prediction results with smaller RMSE than the constant environment model for all the monitoring epochs. Through the evaluation, we can conclude that the proposed approach can provide shorter prediction delay and lower prediction errors than that of the PH model without considering the influence of operational environments.

4.3. Comparison to the constant transition model

To illustrate the performance of the consideration of possible state transitions between monitoring intervals in this research, we compared the proposed model with the constant transition model [32]. In the constant transition model (i.e., Tang's model), the failure rate is described by the PH model, and the values of degradation level are only considered at monitoring points due to the limitation of the periodic inspection policy.

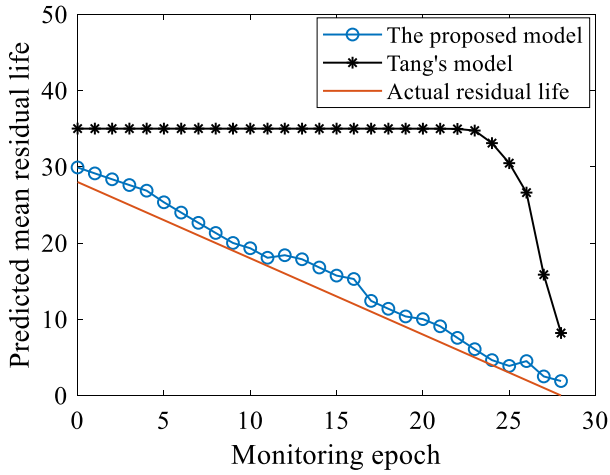
The comparison results are shown in Fig. 10, and it is clear that the proposed model can provide far more accurate prediction than the constant transition model. The prediction residual life of constant transition model has obviously larger prediction error and longer prognostic delay. This is because that the observations are obtained only at monitoring points, and the predicted residual life changes greatly with Weibull distribution, whose hazard rate remains low, and will increase severely in the late stage of the whole life cycle.

To evaluate the proposed model, we have calculated some scientific evaluation indices, including mean error (ME), mean square error (MSE), and mean relative error (MRE) ($MRE = |ARL - MRL| / ARL$, ARL is the actual residual life) of predicted mean residual life for testing the selected failure histories using the above three models. The results are presented in Table 2. Table 2 signifies that for all the selected failure histories, the proposed model provides considerably better performance of MRL prediction with smaller ME, MSE, and MRE than other two PH models.

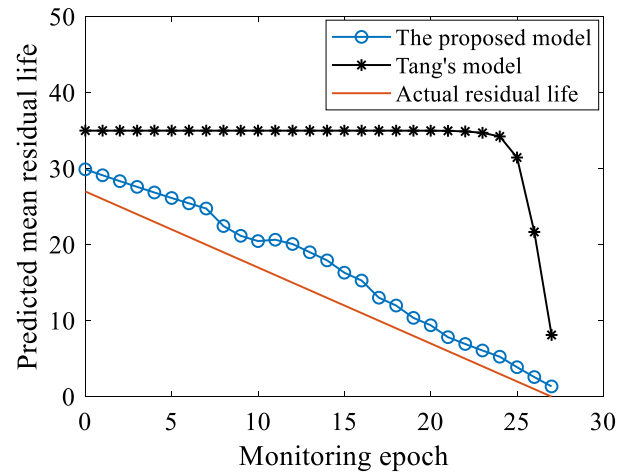
4.4. Comparison to the conventional environment-varying model

We again compare the proposed approach with the conventional environment-varying model in which the environment impact is considered a random process. In Cha's model [33], the external environment is modeled as a Poisson process of shocks and is characterized directly via the hazard rate with deterministic jump on each shock. The shock is considered the occurrence of transition from environment state 1 to state 2 in the proposed model, and the failure rate in the baseline environment (i.e., the low load mode in our assumption) is assumed to be a Weibull distribution.

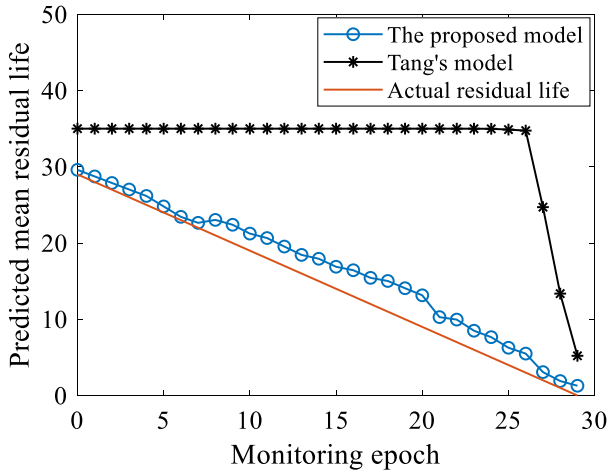
The failure histories #1 and #4 are selected for this comparison, and the results are shown in Fig. 11. The predicted mean residual life evaluated by Cha's model decreases abruptly when



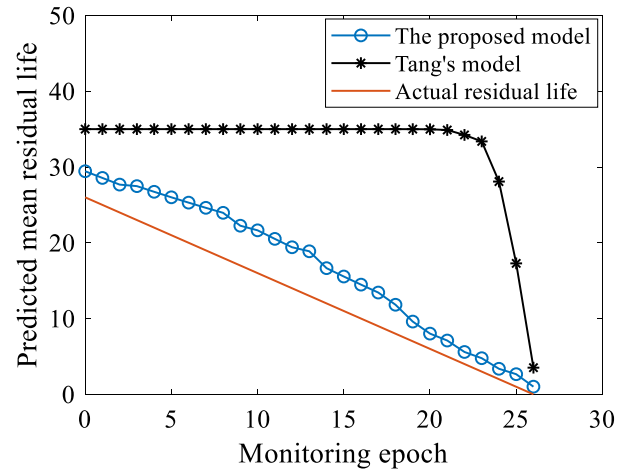
(a) Comparison for data history #1.



(b) Comparison for data history #2.

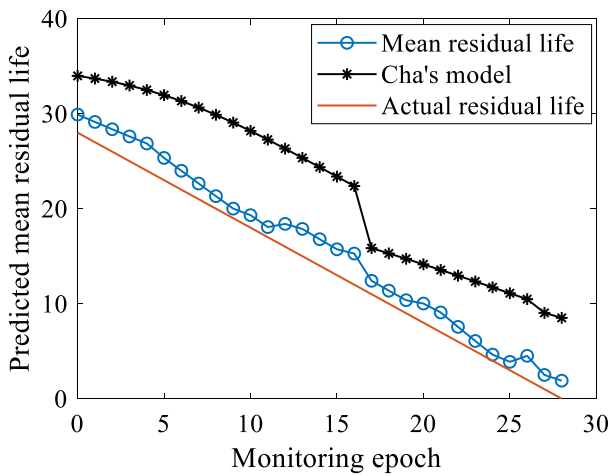


(c) Comparison for data history #3.

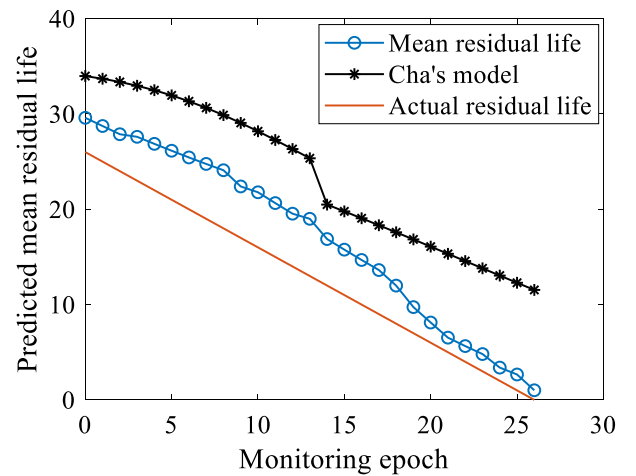


(d) Comparison for data history #4.

Fig. 10. Prediction comparison using non-constant and constant models.



(a) Comparison for history #1



(b) Comparison for history #4

Fig. 11. Prediction comparison using two environment-varying approaches.

external environment transition occurs. However, the proposed model is more applicable for the system operating under continuous environment effects and can provide more accurate prognosis results.

Overall, the proposed approach can be an effective tool to estimate the health characteristics of the gradually degrading engineering systems under varying environments, and three comparison studies demonstrates that our model gives considerably better prediction indicators than the recent developed models, which has demonstrated the superiority of the proposed research.

5. Conclusions

In this paper, a health characteristics prediction method for engineering systems under varying operating environments has been proposed. The internal state of system health is unobservable and partially related to observation process, which is modeled with a dynamic Gamma degradation process affected by aging and varying environments. A novel PH model is presented to link the relation of partial degradation process to system failure rate under varying environments. For health estimation of such systems, a matrix operation-based method is developed to address computation issue of prognostics with non-constant state transitions and dynamic environment profiles. The developed prognostic method can provide comprehensive health characteristics including conditional reliability function, residual life distribution, and mean residual life through basic matrix operations. The closed-form expressions of health characteristics are also obtained. To further illustrate the effectiveness of the proposed method for health characteristics prediction, an example study and comparisons have been developed, which has demonstrated that the proposed approach provides better prognosis with low errors under varying environments.

Our model can be applied to a much wider class of applications. For example, this model can be applied to wind turbine system prognostics and tool wear estimation subject to condition monitoring. The limitation of the proposed approach is that the model cannot be applied for degrading systems under dependent failure modes and a large number of environment states. In future research, the dynamic degradation process which is modeled as Gamma process can be extended to a multi-phase stochastic process with switching models to consider the multiple degradation features under varying environments. In addition, our model can also be extended to the case where the environmental variable is scalar affecting the system degradation. The environmental variable can be modeled as a continuous-state process or general distribution characterized by loads/temperatures.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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